

Group 76 Progress Report:

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1 Introduction

Healthcare providers deal with many vision related illnesses such as DME (Diabetic Macular Edema), CNV (Choroidal Neovascularization), and drusen that impair vision. They utilize retinal scans to detect and determine the conditions. An issue that arises is the uniqueness of each ailment, as each condition varies in subtle ways. Correctly identifying retinal diseases can prove to be a challenge in a plethora of cases and with approximately 30 million OCT scans performed each year, this process can prove time-consuming for healthcare professionals. By having a classification model that can help speed up this process, we can allow healthcare professionals to focus their attention on helping more individuals requiring their aid.

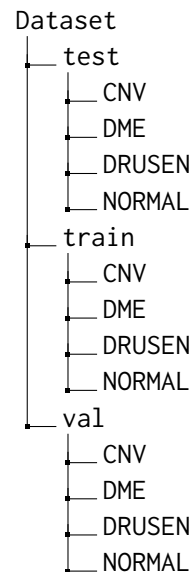
2 Related Work

Here, talk about the related work you encountered for your approach. Cite at least 5 references. Refer to item 2. No one has done exactly your task? Write about the most similar thing you can find. This should be around 0.25-0.5 pages.

3 Dataset

3.1 Folder Structure

When downloading the data, the dataset was partitioned as follows:



3.2 Preprocessing

Each image was loaded from the various folders as a PIL (Pillow, not the Python Imaging Library) Image and then converted to a tensor object. The pipeline varies slightly between *train* when compared to both *test* and *val* (abbreviated to *TV* pipeline for brevity). Both pipelines resized the images to 224 pixels by 224 pixels and normalized it with means [0.485, 0.456, 0.406] and standard deviation (std) of [0.229, 0.224, 0.225]. The values are the means and standard deviations for ImageNet, which is the dataset that ResNet is being trained on. The only difference is that before normalization, the *train* dataset images might be flipped whereas the image data for the *TV* pipeline is not. This allowed for the training set to have a more robust and diverse dataset. However, this is technically not considered an augmentation since we did not create new data (no data distribution changes).

3.3 Annotation

Each image is labeled with an integer to represent its corresponding disease. The images were already named as DISEASE-#####-#.jpeg and located

in their corresponding folders, which allowed for ease of labeling. The numbers for their corresponding disease is shown below:

CNV	0
DME	1
DRUSEN	2
NORMAL	3

4 Features

4.1 Data Input

Since we are doing a computer vision/image classification task, the feature were derived from the pixels of the images. The input for the model was images, which are comprised of pixels. Each pixel is denoted by three values: Red(R), Green(G), and Blue(B). As mentioned previously, each image was cropped and normalized with values corresponding the means and standard deviations for images from ImageNet.

4.2 Features

In terms of selecting the in features, we did not directly do that. This is because it is not required as there are no features to select. Since the pixels are the "feature", this would not be justified. As a result, feature engineering and feature selection was not needed. The same can be said for learning embeddings, as this is not a natural language processing task.

For the out features, there are four main features, which are the four categories of diseases: CNV, DME, DRUSEN, and NORMAL.

5 Implementation

Describe your model and implementation here. Refer to item 4. This may take around a page.

5.1 Learning Model

Instead of training a model from scratch, we used a pretrained neural network and adapted it to our use case. The pretrained neural network selected was ResNet-50, which is trained on the ImageNet dataset. ResNet-50 is a neural network that is 50 layers deep originally developed by Microsoft Research in 2015. This is a type of convolutional neural network (CNN), which is a subset of neural networks that are used for image recognition.

5.2 Implementation Steps

5.2.1 Initializing the Neural Network

To implement the ResNet-50 based neural network, the first step was to instantiate a neural network that uses ResNet-50. This occurs after all the images have been preprocessed. This was called CustomResNet. For the initialization, the input features and classifiers are all defined. The initialization is very similar to typical setup for neural networks.

5.2.2 Freezing and Unfreezing Layers

Upon initialization, the first thing is to freeze all the layers but the last layer. This is because a pretrained model such as ResNet 50 already comes with various useful general features such as edges, textures, and colours in early layers. Since the dataset we use is quite small (four disease classes), training all the layers can result in an increase of time, overfitting, and wiping out pretrained knowledge.

5.2.3 Defining the Criterion and Optimizer

Before compiling the model, two additional items need to be addressed, which are the criterion and optimizer. For the criterion, we used Cross Entropy from the torch.nn (neural network) library. For the optimizer, we used Adam (Adaptive Moment Estimation) from the torch.optim library. The Adam stochastic optimization method uses two key concepts, which are Momentum and Root Mean Square Propagation (RMSprop), combining them to balance the optimization process. The learning rate defined in Adam is 0.001, which is common for machine learning. The optimization is performed only on the parameters of the final layer since the other layers are currently frozen.

5.2.4 Training Data

The dataset has 83484 images in the four folders (CNV, DME, DRUSEN, NORMAL) under training. The distribution is not even, as CNV had significantly more images compared to the other diseases. For this milestone, we took 1000 random photos from each folder of the dataset, resulting in an even spread for each disease. This totaled 4000 images for model training for this milestone. We plan on using all the photos and additional augmentation for the categories with fewer images in the final model to prevent skewing and bias.

5.2.5 Validation and Test Data

Since there are significantly fewer images categorized under these, we used all of them. There are only 32 images for validation spread evenly in the four diseases. The testing data has an even spread too, with 242 images for each disease, totaling 968 images.

5.2.6 Training

For training, we used 10 epochs for the 4000 selected images. We did not do cross validation because of the computation cost that would incur with the large dataset. While this does not seem that intense for the 4000 image dataset used in this milestone, the 80000 data points in the final dataset along with no fewer than 10 epochs would lead to hours of computation.

5.2.7 Testing

6 Results and Evaluation

How are you evaluating your model? What results do you have so far? What are your baselines? Refer to item 5. This may take around 0.5 pages.

7 Feedback and Plans

Write about your plans for the remainder of the project. This should include a discussion of the feedback you received from your TA, and how you plan to improve your approach. Reflect on your implementation and areas for improvement. Refer to item 6. This may be around 0.5 pages.

8 Template Notes

You can remove this section or comment it out, as it only contains instructions for how to use this template. You may use subsections in your document as you find appropriate.

8.1 Tables and figures

See Table 1 for an example of a table and its caption. See Figure ?? for an example of a figure and its caption.

8.2 Citations

Table 1 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\cite{t}` (cite in text) to get “author (year)” citations, like this citation to a paper by ?. You can use the command `\citep` (cite in parentheses) to get “(author, year)” citations (?). You can use the command `\citealp` (alternative

cite without parentheses) to get “author, year” citations, which is useful for using citations within parentheses (e.g. ?).

8.3 References

Many websites where you can find academic papers also allow you to export a bib file for citation or bib formatted entry. Copy this into the custom.bib and you will be able to cite the paper in the $\text{\LaTeX}$. You can remove the example entries.

8.4 Equations

An example equation is shown below:

$$A = \pi r^2 \tag{1}$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command. This an example cross-reference to Equation 1. You can also write equations inline, like this: $A = \pi r^2$.

Team Contributions

Write in this section a few sentences describing the contributions of each team member. What did each member work on? Refer to item 7.

Output	natbib command	ACL only command
(?)	\citep	
?	\citealp	
?	\citet	
(?)	\citeyearpar	
?’s (?)		\citeposs

Table 1: Citation commands supported by the style file.